

IRUMA AB, Japan

WMO: 476430

Lat: 35.842N

Lon: 139.411E

Elev: 305

StdP: 14.53

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.5	28.3	5.0	7.2	39.2	8.3	8.5	38.8	25.2	43.8	22.9	43.5	3.6	270	0.417

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.5	94.6	77.7	91.6	76.9	89.4	76.3	79.5	88.9	78.6	87.4	77.6	85.8	9.2	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.2	142.9	83.0	75.6	135.7	82.2	75.3	134.0	81.9	43.1	89.3	42.1	88.1	41.2	84.8	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.5	19.3	16.7	21.1	98.3	2.0	2.0	19.7	99.7	18.5	100.9	17.4	102.0	16.0	103.5		
			DB	WB	18.3	81.7	1.7	1.6	17.1	82.9	16.2	83.8	15.2	84.7	14.0	85.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.4	38.9	40.6	46.7	56.6	65.4	71.2	78.8	80.8	73.7	63.3	52.4	43.2
	DBStd	15.33	3.87	4.63	5.89	6.36	5.17	4.93	5.74	4.98	5.90	5.35	5.32	4.81
	HDD50	1020	344	267	140	17	0	0	0	0	0	35	217	
	HDD65	3539	808	684	568	260	57	8	0	6	96	377	675	
	CDD50	4458	1	3	37	213	478	637	894	956	711	413	108	7
	CDD65	1501	0	0	0	6	70	195	429	491	266	44	0	0
	CDH74	12518	0	0	2	66	457	1106	3977	4918	1814	177	1	0
	CDH80	4660	0	0	0	8	83	281	1643	2060	557	28	0	0
Wind	WSAvg	6.5	7.0	7.6	8.0	7.9	7.0	6.0	5.8	6.0	5.8	5.7	5.6	6.2
Precipitation	PrecAvg	58.9	2.0	1.9	3.3	4.1	5.3	6.7	6.7	7.0	8.9	7.6	3.2	2.2
	PrecMax	82.2	4.9	3.7	6.3	10.2	10.5	15.5	12.8	15.8	15.9	24.4	11.5	6.3
	PrecMin	41.2	0.2	0.1	1.1	1.2	2.2	2.8	2.7	1.7	1.5	0.9	0.1	0.4
	PrecStd	8.2	1.4	1.1	1.3	2.2	2.1	2.6	3.1	4.0	4.3	5.3	2.2	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.5	64.3	71.5	80.3	86.3	91.3	97.0	97.1	92.9	84.1	71.4	62.7
		MCWB	45.4	50.9	54.8	61.3	65.7	73.5	78.2	78.2	75.6	70.0	58.0	51.9
	2%	DB	53.5	57.2	66.2	75.3	82.2	86.2	94.6	94.9	89.4	78.7	67.8	58.7
		MCWB	42.4	45.4	52.0	59.0	64.1	71.5	77.8	78.0	75.3	66.1	57.0	47.3
	5%	DB	50.3	53.5	62.4	71.7	78.9	82.8	91.5	92.8	86.0	75.2	64.5	55.4
		MCWB	39.7	42.8	50.1	57.3	62.9	69.5	77.1	77.6	74.1	64.2	54.5	45.6
	10%	DB	48.2	50.1	58.6	68.2	75.6	80.5	89.2	89.7	82.7	71.9	62.2	52.2
		MCWB	38.5	40.2	47.9	56.1	61.8	68.4	76.6	76.9	72.0	62.4	53.9	43.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.6	53.2	57.8	66.5	70.9	77.0	80.9	81.0	78.8	72.4	62.6	54.2
		MCDB	53.3	63.6	66.1	73.8	78.0	86.1	91.2	90.8	86.7	79.1	67.3	58.9
	2%	WB	44.2	47.3	54.6	62.4	67.7	74.9	79.4	79.8	77.2	69.1	59.9	50.4
		MCDB	50.5	54.5	63.4	70.8	75.5	82.2	89.0	89.3	84.4	75.5	64.4	54.6
	5%	WB	41.5	44.2	51.7	59.9	65.7	72.9	78.4	78.9	76.0	66.9	57.4	47.4
		MCDB	48.1	50.2	59.9	68.4	74.1	79.3	87.4	87.9	82.6	72.6	62.1	52.8
	10%	WB	39.3	41.6	48.9	57.8	64.4	71.1	77.3	78.1	74.7	65.0	55.4	45.0
		MCDB	46.4	48.3	56.6	65.8	71.6	76.9	85.6	86.5	80.7	70.4	59.5	50.5
Mean Daily Temperature Range	5% DB	MDBR	18.0	17.8	18.4	18.9	17.2	13.8	14.0	14.5	13.4	13.7	15.8	17.4
		MCDBR	21.3	23.1	24.8	24.2	22.5	18.8	18.5	18.5	17.3	17.4	19.4	20.3
		MCWBR	13.2	14.4	14.6	12.4	9.8	7.8	6.6	6.1	7.1	9.0	11.2	12.5
	5% WB	MCDBR	19.5	20.4	22.1	20.1	18.1	14.5	16.7	16.3	15.2	14.3	15.3	17.0
	MCWBR	13.2	14.0	14.6	12.0	9.5	7.3	6.6	6.5	7.1	8.4	10.2	11.8	
Clear-Sky Solar Irradiance	taub	0.371	0.419	0.487	0.541	0.570	0.613	0.603	0.597	0.524	0.484	0.417	0.373	
	taud	2.248	2.110	1.950	1.865	1.849	1.812	1.887	1.888	2.049	2.085	2.194	2.274	
	Ebn at Noon	255	254	247	240	235	225	226	224	234	230	235	246	
	Edh at Noon	34	43	56	63	65	68	62	61	50	44	36	31	
All-Sky Solar Radiation	RadAvg	875	1056	1314	1526	1637	1442	1524	1526	1204	976	809	765	
	RadStd	57	92	80	144	171	129	250	150	107	102	75	56	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.83	+0.99	N/A	N/A	N/A	N/A	-92	-200	+254	N/A
(47)	Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page